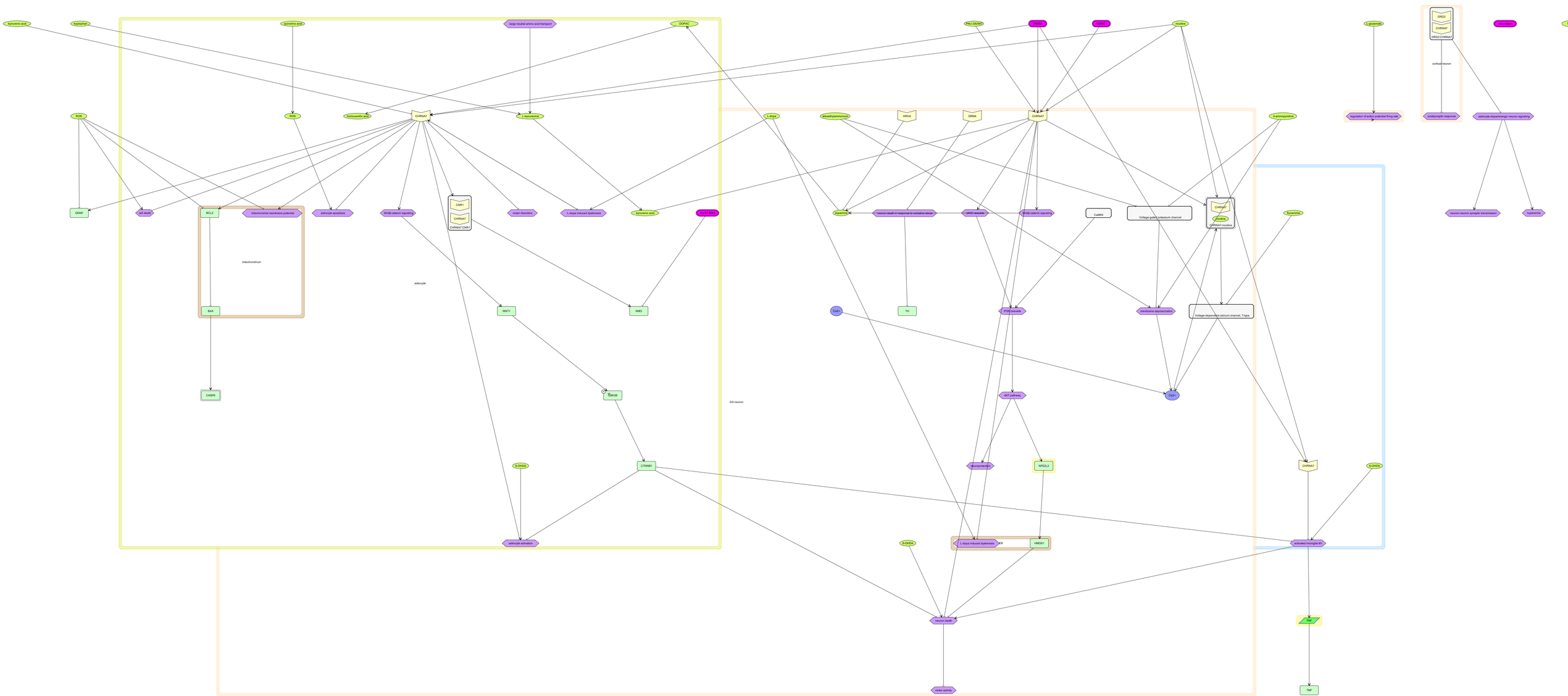

ROS
GDNF
cell death
mitochondrial membrane potential
ROS
BCL2
BAX
mitochondrium
CASP9
CASP9
CHRNA7
nicotine
DMXBA
nicotine
Wnt/β-catenin signaling
WNT1
P
S9
GSK3B
astrocyte activation
6-OHDA
CTNNB1
astrocyte
CHRNA7
6-OHDA
activated microglia M1
activated microglia M1
nucleus
TNF
TNF
TNF
microglia
kynurenic acid
CHRNA7
CNR1
CHRNA7
CHRNA7:CNR1
rimonabant
CNR1
THC
KMO
Ro 61-8048
L-glutamate
L-glutamate
astrocyte
AQW051
L-dopa
CHRNA7
DMXBA
CHRNA7
CHRNA7
PNU-282987
CHRNA7
neuron death
L-dopa induced dyskinesia
6-OHDA
HMOX1
ER
motor activity
JAK2 cascade
PI3K cascade
NFE2L2
AKT pathway
nucleus
regulation of action potential firing rate
glutamatergic pyramidal neuron
postsynaptic response
DRD2
CHRNA7
DRD2:CHRNA7
neuron-neuron synaptic transmission
astrocyte-dopaminergic neuron signaling
hyperemia
cortical neuron
DRD2
CHRNA7
regulation of action potential firing rate
glutamatergic pyramidal neuron
Ca2+
Ca2+
tetraethylammonium
membrane depolarization
4-aminopyridine
Voltage-gated potassium channel
CHRNA7
nicotine
&
CaMKII
PI3K cascade
AKT pathway
neuroprotection
CHRNA7
nicotine
CHRNA7:nicotine
flunarizine
Voltage-dependent calcium channel, T-type
Ca2+
Ca2+
GDNF
nicotine
neuron death in response to oxidative stress
GRM4
CHRNA7
TH
HRH4
quinolinic acid
ROS
GDNF
astrocyte apoptosis
motor disorders
dopamine
dopamine
DOPAC
homovanillic acid
Wnt/β-catenin signaling
large neutral amino acid transport
L-kynurenine
L-kynurenine
tryptophan
kynurenic acid
L-dopa induced dyskinesia
L-dopa
CHRNA7
DA neuron
nicotine
astrocyte
regulation of action potential firing rate
glutamatergic pyramidal neuron

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